

The Himalayan Tightrope

Balancing the Economy-Environment Trade-off

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The Trade-off Exists...

- Infrastructure growth is helpful for
- There is heavy infra investment taking place in the Himalayas.

The Trade-off Exists. . .

- Infrastructure growth is helpful for
- There is heavy infra investment taking place in the Himalayas.
- But then, there is a growing *long-run* and *human* cost.

... But Do We Measure It Right?

- The Environmental Kuznets Curve (EKC)
- The IRAM Models
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- How does *infrastructure-led* growth relate to environmental degradation?
- Does the Environmental Kuznets curve hold?
- Do macro-level models hold for:
 - sensitive ecologies?
 - micro-level data?

Our answer: Not well.

- We build a high-res geospatial dataset at village/town level
- We estimate the EKC: 5 indicators, 20 years of data
- We focus on two Himalayan states, where incentives exist for both infrastructure investment and ecological sustainability

Data

Unit Our basic unit is the *shrid*, from the SHRUG v2, Development Data Lab.

Econ

Night lights

Why? Strong correlation with GDP growth even at subnational level (Doll, Muller, and Morley, 2006).

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Built-up area

Why? Detects economic activity at highly local levels (Baragwanath et al., 2021).

Source ESA, Copernicus GHSL product, five-year intervals.

PM2.5 air quality

Why? Respected measure of environmental degradation (Caviglia-Harris, Chambers, and Kahn, 2009).

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Land surface temperature

Why? Long-run changes from environmental degradation, global warming (European Space Agency, n.d.).

Source NASA MODIS, yearly aggregates from 8-weekly data.

Forest cover

Why? Relevant for infra growth, also used for horizontal devolution (Wang, 2003; XV Finance Commission, 2020).

Source SHRUG v2, Development Data Lab, Terra MODIS (NASA).

Model

Model 1: Entire Time Period

$$\text{Econ}_{it} = \beta_0 + \beta_1(\text{Env}_{it}) + \beta_1(\text{Env}_{it}^2) + \eta_i + \gamma_t + \epsilon_{it}$$

where:

- Econ is the economic index variable.
- Env is the environmental index variable.
- η_i are *shrid*-level fixed effects.
- γ_i are year fixed effects.
- ϵ_{it} is the idiosyncratic error term.

- Investigates the trade-off between economic growth and environmental sustainability.
- Uses high-resolution geospatial data for Himachal Pradesh and Uttarakhand.
- Develops a novel dataset of economic and environmental indicators over two decades.

Context: Why Build Infrastructure?

- Economic, political, and administrative motivations.
- Border security concerns with China, Nepal, and Bhutan.
- Economic transition from agriculture to manufacturing and services.
- Role of infrastructure in connectivity, digital access, and energy generation.

Theoretical Framework: Environmental Kuznets Curve (EKC)

- Hypothesis: Inverted-U relationship between economic growth and environmental degradation.
- Early stages: Growth leads to increased degradation.
- Later stages: Growth enables sustainability efforts.
- Contradictory findings in recent literature.

- Study unit: SHRUG village/town-level data.
- Economic indicators: Night-time lights, built-up area.
- Environmental indicators: Air quality (PM2.5), forest cover, land surface temperature.
- Data sources: NASA MODIS, GHSL, VIIRS, DMSP-OLS.

Findings: Temporal Shifts in Himachal Pradesh

- Early 2000s - Mid 2000s: Forest cover increases in northern districts, declines in southern regions.
- 2010s: Growth in night-time lights, economic activity expands.

Findings: Temporal Shifts in Uttarakhand

- Early 2000s - Mid 2000s: Forest cover increases in the north, declines in the south.
- Mid 2000s - 2010s: Widespread forest cover loss.
- Post-2010: Recovery in forest cover.
- Night-time lights show growth across all periods.

- Quadratic fixed-effects model:
- Economic Index (NTL, built-up area) vs. Environmental Index (PM2.5, forest cover, LST).
- Fixed effects used based on Hausman test.

Regression Results: First Epoch

- Estimated relationship follows the EKC (inverted U-shape).
- Economic growth initially leads to environmental degradation, then stabilizes.

Regression Results: Second Epoch

- Relationship reverses to a U-shape.
- Economic growth initially reduces degradation but later exacerbates it.

- Results suggest heterogeneity across time.
- EKC not stable in ecologically sensitive regions.
- Potential tipping point in Himalayan environmental dynamics.

- Need for localized, dynamic policy frameworks.
- Infrastructure planning should integrate ecological risk assessments.
- Emphasis on renewable energy, sustainable urban planning.

- Economy-environment trade-offs in the Himalayas are complex.
- EKC model does not universally apply.
- Future research: pinpoint specific projects to build a treatment group for causal analysis.

Thank you.

Open to questions/comments.